

Subject Description Form

Subject Code	EEE508
Subject Title	VLSI Technology and Design
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This subject aims to familiarize students with CMOS integrated technology and equip them with fundamental know-how on which they will build their future careers in the semiconductor foundries and research areas. The students will not only understand the basic technology and process flow for the fabrication of integrated CMOS VLSI circuits but also various aspects of VLSI system design.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: Category A: Professional/academic knowledge and skills - Understand of VLSI design methodology, VLSI design flow, and design hierarchy. - Understand the VLSI circuits design, analyze circuit performance, and analyze logic circuits of CMOS - Learn IC technologies Category B: Attributes for all-roundedness - Communicate effectively. - Think critically and creatively. - Assimilate new technological development in related field
Subject Synopsis/ Indicative Syllabus	Introduction: Overview of fundamental concepts of VLSI and MOS transistor principle and theory VLSI technology: Historical perspective, technology development milestones, industry and trends today. Introduction to design methodologies: The VLSI design problem, the VLSI design methods and technologies, layout methodologies, etc. Device modelling and design: SPICE modelling, CMOS logic circuit design, parameter extraction and PCB design. Microchip fabrication techniques: Process flow of basic ICs, design rule, layout, VLSI process integration, 3D integration, packaging and assembly, multilayer PCB Semiconductor device packaging: Materials and processes used to semiconductor device packaging, stresses induced due to packaging and failure analysis techniques.

Teaching/Learning Methodology	The introduction of VLSI and MOS transistor theory will be described and explained in lectures. Various VLSI technologies will be presented in lectures. Students will also be required to study the failure analyses, which play an important role in process development.																																																					
	Teaching/Learning Methodology		Intended subject learning outcomes																																																			
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	Lectures		√	√	√	√	√	√																																														
	Case study and presentation		√	√	√	√	√	√																																														
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><td rowspan="2">Specific assessment methods/tasks</td><td rowspan="2">% weighting</td><td colspan="6">Intended subject learning outcomes to be assessed (Please tick as appropriate)</td></tr><tr><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td><td>f</td></tr><tr><td>1. Assignments</td><td>30%</td><td>√</td><td>√</td><td></td><td>√</td><td>√</td><td>√</td></tr><tr><td>2. Presentation</td><td>20%</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>3. Examination</td><td>50%</td><td>√</td><td>√</td><td>√</td><td></td><td>√</td><td>√</td></tr><tr><td>Total</td><td>100 %</td><td colspan="6"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Assignments: Let students review the taught materials, do further reading for deeper learning, and understand better the taught knowledge. Students may find these readings useful. Presentation: Presentation is designated based on their self-readings and share their ideas and views about the modern VLSI technologies through discussions. Mutual discussions are encouraged in order to summarize the findings in a presentation. Examination: The examination will evaluate student’s understanding and knowledge they have achieved in the intended learning outcomes.								Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)						a	b	c	d	e	f	1. Assignments	30%	√	√		√	√	√	2. Presentation	20%	√	√	√	√	√	√	3. Examination	50%	√	√	√		√	√	Total	100 %						
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Total	100 %																																																					
Student Study Effort Expected	Class contact:																																																					
	▪ Lectures						24 Hrs.																																															
	▪ Tutorial, case study and presentation						15 Hrs.																																															
	Other student study effort:																																																					
	▪ Further reading, doing homework/assignments and preparing for the subject						81 Hrs.																																															
	Total student study effort						120 Hrs.																																															

Reading List and References	Textbook • Neil Weste, David Harris, CMOS VLSI Design: A Circuits and Systems Perspective 4th Edition, Pearson, 2010. References • John P. Uyemura, VLSI Design Principles, John Wiley, 2002. • E. Fabricious, Introduction to VLSI design, McGraw-Hill, 1990. • Wayne Wolf, Modern VLSI Design, Pearson Education, 2003.
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